

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management
Systems COURSE CODE: CSA0503

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.*
(BL2, BL3)

Activity Tool : Case Study (Medium)
Assessment Tool Weightage: 30%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.	BL2 – Understand	5
2	A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.	BL3 – Apply	5
3	Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.	BL3 – Apply	5
4	Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.	BL3 – Apply	5
5	Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.	BL2 – Understand	5
6	Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.	BL3 – Apply	5
7	Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.	BL2 – Understand	5

8	A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.	BL3 – Apply	5
9	Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.	BL3 – Apply	5
10	Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.	BL3 – Apply	5

Code of Conduct:

I K.MOHAMMED NAFEEZ[192571070] certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:K.MOHAMMED NAFEEZ

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 – Exemplar y	Level 3 – Proficien t	Level 2 – Developin g	Level 1 – Beginnin g
Understandi ng of Case	25	Thorough understand i ng; clearly identifies all key issues	Good understandi ng with most issues identified	Basic understandi ng; some issues missed	Poor understand i ng; problem not identified correctly
Applicatio n of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriat e concepts with minor gaps	Limited application of concepts	Fails to apply relevant concept s

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions	Logical solutions with	Basic solutions with weak	Incorrect or no solution provided

		with strong justification	adequate justification	justification	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANSWERS:

1. Online Shopping – 3NF

START

Identify FDs:

OrderID → CustID

CustID → CustName

ProdID → ProdName, Price

(OrderID, ProdID) → Qty

Create Customer(CustID, CustName)

Create Product(ProdID, ProdName, Price)

Create Orders(OrderID, CustID)

Create OrderDetails(OrderID, ProdID, Qty)

```
mysql> SELECT O.OrderID, C.CustName, P.ProdName, OD.Qty, P.Price
-> FROM Orders O
-> JOIN Customer C ON O.CustID=C.CustID
-> JOIN OrderDetails OD ON O.OrderID=OD.OrderID
-> JOIN Product P ON OD.ProdID=P.ProdID;
+-----+-----+-----+-----+-----+
| OrderID | CustName | ProdName | Qty | Price |
+-----+-----+-----+-----+-----+
|      1 | Ravi    | Laptop  | 1   | 50000 |
|      1 | Ravi    | Mouse   | 2   | 800   |
+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

END

2. Hospital – BCNF

START

Identify FDs:

PID → PName, DocID, WardNo

DocID → DocName, Specialization

WardNo → WardName

Create Patient(PID, PName, DocID, WardNo)

Create Doctor(DocID, DocName, Specialization)

Create Ward(WardNo, WardName)

```
mysql> SELECT P.PID, P.PName, D.DocName, D.Specialization, W.WardName
-> FROM Patient P
-> JOIN Doctor D ON P.DocID=D.DocID
-> JOIN Ward W ON P.WardNo=W.WardNo;
```

```
+-----+-----+-----+-----+-----+
| PID | PName | DocName | Specialization | WardName |
+-----+-----+-----+-----+-----+
| 101 | Ravi  | Kumar   | Cardiology     | General  |
| 102 | Sita  | Priya   | Neurology      | ICU      |
+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

END

3. University Enrollment – 2NF

START

Identify FDs:

$SID \rightarrow SName$

$CourseID \rightarrow CourseName, Faculty$

$(SID, CourseID) \rightarrow Grade$

Create Student(SID, SName)

Create Course(CourseID, CourseName, Faculty)

Create Enrollment(SID, CourseID, Grade)

END

```
mysql> SELECT E.SID, S.SName, C.CID, C.CName, C.Faculty, E.Grade
-> FROM Enrollment E
-> JOIN Student S ON E.SID=S.SID
-> JOIN Course C ON E.CID=C.CID;
```

```
+-----+-----+-----+-----+-----+-----+
| SID | SName | CID | CName          | Faculty | Grade |
+-----+-----+-----+-----+-----+-----+
| 101 | Arun  | 201 | DBMS           | Kumar   | A      |
| 101 | Arun  | 202 | Operating Systems | Ravi    | B      |
| 102 | Priya | 201 | DBMS           | Kumar   | A      |
| 102 | Priya | 202 | Operating Systems | Ravi    | A      |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

4. Airline Booking – BCNF

START

Identify FDs:

$FlightNo \rightarrow DeptCity, ArrCity$

$PassID \rightarrow PassName$

$(FlightNo, Date, Seat) \rightarrow PassID$

Create Flight(FlightNo, DeptCity, ArrCity)

Create Passenger(PassID, PassName)

Create Booking(FlightNo, PassID, Seat, Date)

END

```
mysql> SELECT B.FlightNo, B.PassID, P.PassName, B.Seat,
-> F.DeptCity, F.ArrCity, B.BookingDate
-> FROM Booking B
-> JOIN Passenger P ON B.PassID=P.PassID
-> JOIN Flight F ON B.FlightNo=F.FlightNo;
+-----+-----+-----+-----+-----+-----+-----+
| FlightNo | PassID | PassName | Seat | DeptCity | ArrCity | BookingDate |
+-----+-----+-----+-----+-----+-----+-----+
| AI101    | 1      | Ravi     | 12A  | Chennai  | Delhi   | 2026-08-20   |
| AI102    | 2      | Kiran    | 10B  | Mumbai   | Delhi   | 2026-08-21   |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

5. HR System – 3NF

START

Identify FDs:

EmpID → EmpName, DeptID, JobID

DeptID → DeptName, Location

JobID → JobTitle, SalaryGrade

Create Employee(EmpID, EmpName, DeptID, JobID)

Create Department(DeptID, DeptName, Location)

Create Job(JobID, JobTitle, SalaryGrade)

END

```
mysql> SELECT E.EmpID, E.EmpName, D.DeptName, D.Location,
-> J.JobTitle, J.SalaryGrade
-> FROM Employee E
-> JOIN Department D ON E.DeptID=D.DeptID
-> JOIN Job J ON E.JobID=J.JobID;
+-----+-----+-----+-----+-----+-----+
| EmpID | EmpName | DeptName | Location | JobTitle           | SalaryGrade |
+-----+-----+-----+-----+-----+-----+
| 1001  | Ravi    | IT       | Chennai  | Software Engineer  | G5          |
| 1002  | Sita    | HR       | Bangalore | HR Manager         | G7          |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

6. Library – 4NF

START

Identify MVDs:

BookID →→ Author

BookID →→ Genre

Create Book(BookID, Title)

Create BookAuthor(BookID, Author)

Create BookGenre(BookID, Genre)

Create Member(MemberID, MemberName)

Create Borrow(MemberID, BookID, BorrowDate)

END

```
mysql> SELECT B.BookID, B.Title, A.Author, G.Genre
-> FROM Book B
-> JOIN BookAuthor A ON B.BookID=A.BookID
-> JOIN BookGenre G ON B.BookID=G.BookID;
+-----+-----+-----+-----+
| BookID | Title           | Author       | Genre       |
+-----+-----+-----+-----+
| 101    | Database Systems | Korth        | Database    |
| 101    | Database Systems | Korth        | Education   |
| 101    | Database Systems | Ramakrishnan | Database    |
| 101    | Database Systems | Ramakrishnan | Education   |
+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

7. Telecom Billing

START

Identify candidate keys:

CustomerID

Phone

BillID

Identify FDs:

CustomerID → CustomerName, Phone

Phone → CustomerID, CustomerName

BillID → CustomerID, Amount

Create Customer(CustomerID, CustomerName, Phone)

Create Bill(BillID, CustomerID, Amount)

END

```
mysql> SELECT B.BillID, C.CustomerID, C.CustomerName,
-> C.Phone, B.Amount
-> FROM Bill B JOIN Customer C ON B.CustomerID=C.CustomerID;
+-----+-----+-----+-----+-----+
| BillID | CustomerID | CustomerName | Phone       | Amount |
+-----+-----+-----+-----+-----+
| 1001   | 101        | Ravi         | 9876543210 | 599    |
| 1002   | 102        | Kiran        | 9876543211 | 299    |
+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

8. Retail Inventory – Normalization

START

Identify FDs:

PID → PName, SupplierID, Category, Price

SupplierID → SupplierName

Create Product(PID, PName, SupplierID, Category, Price)
 Create Supplier(SupplierID, SupplierName)
 Create Warehouse(WarehouseID, WarehouseName)

END

```
mysql> SELECT P.PID, P.PName, S.SupplierName,
-> P.Category, P.Price, P.Warehouse
-> FROM Product P JOIN Supplier S ON P.SupplierID=S.SupplierID;
```

PID	PName	SupplierName	Category	Price	Warehouse
501	Laptop	ABC	Electronics	50000	Chennai
502	Mouse	XYZ	Accessories	800	Chennai

2 rows in set (0.01 sec)

9. School ERP – BCNF

START

Identify FDs:

StudentID → StudentName, DeptID

DeptID → DeptName

SubjectID → SubjectName, TeacherID

TeacherID → TeacherName

Create Student(StudentID, StudentName, DeptID)

Create Department(DeptID, DeptName)

Create Subject(SubjectID, SubjectName, TeacherID)

Create Teacher(TeacherID, TeacherName)

Create StudentSubject(StudentID, SubjectID, Marks)

END

```
mysql> SELECT S.StudentID, S.StudentName, D.DeptName,
-> SU.SubjectName, T.TeacherName, SS.Marks
-> FROM StudentSubject SS
-> JOIN Student S ON SS.StudentID=S.StudentID
-> JOIN Department D ON S.DeptID=D.DeptID
-> JOIN Subject SU ON SS.SubjectID=SU.SubjectID
-> JOIN Teacher T ON SU.TeacherID=T.TeacherID;
```

StudentID	StudentName	DeptName	SubjectName	TeacherName	Marks
1	Ravi	Computer Science	DBMS	Kumar	85
1	Ravi	Computer Science	Operating Systems	Priya	90
2	Kiran	Computer Science	DBMS	Kumar	78

3 rows in set (0.01 sec)

10. Movie Streaming – 5NF

START

Identify Join Dependency:

Movie, Actor, Platform

```
Create Movie(MovieID, MovieName)
Create Actor(ActorID, ActorName)
Create Platform(PlatformID, PlatformName)
Create MovieActor(MovieID, ActorID)
Create MoviePlatform(MovieID, PlatformID)
Create ActorPlatform(ActorID, PlatformID)
```

Verify lossless join

END

```
mysql> SELECT M.MovieName, A.ActorName, P.PlatformName
-> FROM Movie M
-> JOIN MovieActor MA ON M.MovieID=MA.MovieID
-> JOIN Actor A ON MA.ActorID=A.ActorID
-> JOIN MoviePlatform MP ON M.MovieID=MP.MovieID
-> JOIN Platform P ON MP.PlatformID=P.PlatformID;

+-----+-----+-----+
| MovieName | ActorName | PlatformName |
+-----+-----+-----+
| RRR       | NTR       | Netflix      |
| RRR       | Ram Charan | Netflix      |
| RRR       | NTR       | Prime Video  |
+-----+-----+-----+
3 rows in set (0.01 sec)
```